

Company: NVIDIA
Role: AI Engineer
Difficulty: Hard
Round: Technical Round

Overall Experience:

Technical Round 1: Implement a convolution operation from scratch in Python.
Detailed discussion on backpropagation mathematics, gradient descent variants,
and vanishing gradient solutions (ReLU, batch normalization, residual connections).

Technical Round 2: Explain attention mechanism in Transformers.
Implement beam search decoding.
Discuss GPU vs CPU memory bandwidth bottlenecks and
Tensor Core optimization for deep learning workloads.

Tips: NVIDIA expects deep ML theory combined with hardware/systems understanding.
Read research papers weekly and implement them from scratch.